

PRD — Team Task Management System

1. Product Overview

Product Name: Team Task Management System

A web-based application that allows users to create or join organizations, manage projects, assign team members, and track tasks with **role-based access control**.

2. Problem Statement

Teams need a centralized system to:

- Organize members into organizations
 - Create and manage projects
 - Assign members to projects
 - Create and assign tasks
 - Control what admins and members can access
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3. User Roles

Admin

Can:

- Manage organization
- Manage organization members
- Create/manage projects
- Assign project members
- Create, assign and delete tasks

Member

Can:

- Join organizations
- View organizations they belong to

- View assigned projects
- View project members
- View and update tasks within assigned projects

Role is **organization-specific**, so the same user can be an admin in one organization and a member in another.

4. Core Features

Authentication

- Register
- Login
- JWT access token
- Refresh token
- Logout
- Logout from all sessions

Organization Management

- Create organization
- Generate unique invite code
- Join organization using invite code
- View user's organizations
- View organization details
- Update organization
- Delete organization

Member Management

- View organization members
- Change member role
- Remove member

Project Management

- Create project
- View projects
- View project details
- Update project
- Delete project

Project Members

- Assign members to project
- View project members
- Remove project members

Task Management

- Create task
- View project tasks
- View individual task
- Update task
- Assign/reassign task
- Delete task

5. User Flow

Register / Login



Dashboard



Create Organization

OR

Join Organization

↓
Select Organization
↓
Manage Projects
↓
Assign Project Members
↓
Create & Assign Tasks
↓
Track / Update Tasks

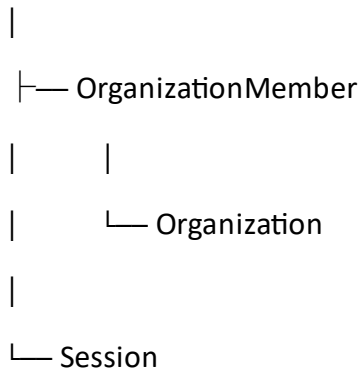
6. Authorization Rules

	Admin	Member
Create Project	✓	X
Update Project	✓	X
Delete Project	✓	X
Assign Project Member	✓	X
Create Task	✓	X
View Tasks	✓	✓
Update Task	✓	✓
Assign/Reassign Task	✓	X
Delete Task	✓	X
Manage Org Members	✓	X

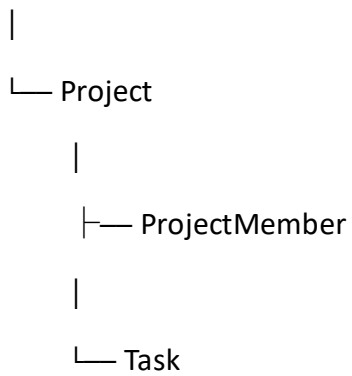
Access to projects/tasks is also based on **membership**, not simply on knowing an ID.

7. Database Entities

User



Organization



Main models:

- User
- Organization
- OrganizationMember
- Project
- ProjectMember
- Task
- Session

8. API Modules

/api/auth
/api/organizations
/api/organizations/:organizationId/members
/api/organizations/:organizationId/projects
/api/projects/:projectId
/api/projects/:projectId/members
/api/projects/:projectId/tasks
/api/tasks/:taskId

9. Technology Stack

Frontend

- React
- Vite
- Tailwind CSS

Backend

- Node.js
- Express.js
- JWT
- bcrypt

Database

- MongoDB
- Mongoose

Development

- Postman
- Git
- GitHub

10. Non-Functional Requirements

- Secure authentication
- Password hashing
- Protected API routes
- Role-based authorization
- Input validation
- Proper HTTP status codes
- Error handling
- Scalable backend structure
- Responsive frontend

11. Success Criteria

The project is considered complete when:

1. User can register/login securely.
2. User can create or join multiple organizations.
3. Organization creator automatically becomes admin.
4. Admin can manage organization members.
5. Admin can create projects.
6. Admin can assign project members.
7. Admin can create and assign tasks.
8. Members can access only their assigned projects/tasks.
9. Authentication and authorization work across protected APIs.
10. Frontend successfully integrates with the backend.

One-line Product Definition

A role-based team collaboration platform where organizations manage projects, project members, and tasks through secure authentication and authorization.